A Survey on Secure Task Scheduling for Digital Twin-Enabled Edge–Cloud Networks

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ABSTRACT

Edge–cloud computing combines the proximity of edge resources with the elasticity of cloud infrastructure, but the resulting environment introduces heterogeneous resources, dynamic workloads, variable network conditions, and security-sensitive data. Task scheduling therefore extends beyond minimizing latency or energy consumption and must account for privacy, trust, risk, and service deadlines. Digital Twin (DT) technology provides a continuously updated virtual representation of computing and communication resources that can support monitoring, prediction, and scheduling. Recent literature shows growing use of reinforcement learning, deep reinforcement learning, metaheuristics, and multi-objective optimization for computation offloading and resource allocation. At the same time, privacy-aware and security-aware offloading studies demonstrate that untrusted edge resources and information exposure can materially alter scheduling decisions. This survey organizes these developments into traditional, optimization-based, AI-based, privacy-aware, security-aware, DT-based, multi-objective, and hybrid approaches. The reviewed literature indicates a recurring separation between performance optimization and security/risk management, although several recent studies combine subsets of these concerns. Based on these observations, a potential research direction is an adaptive Local–MEC–Cloud scheduler in which a Digital Twin supplies current state information, an AI model estimates security risk, and a Genetic Algorithm performs multi-objective resource selection. The direction also incorporates adaptive re-scheduling and controlled security-event simulation. Evaluation should jointly examine latency, energy, resource utilization, deadline satisfaction, scheduling overhead, security errors, privacy exposure, and recovery behavior.

KEYWORDS

Digital Twin; Edge Computing; Mobile Edge Computing; Cloud Computing; Task Scheduling; Genetic Algorithm; Cybersecurity; Privacy Preservation; AI-Based Risk Assessment; Secure Offloading; Resource Optimization; Edge–Cloud Networks

1. INTRODUCTION

Cloud-centric computing provides scalable processing and storage, but applications with strict latency, mobility, and data-locality requirements can experience communication delay and dependence on remote infrastructure. Edge computing addresses part of this problem by placing computation closer to data sources, while Mobile Edge Computing (MEC) places computing

capabilities near mobile users and access networks [1], [2]. The resulting edge–cloud continuum allows a task to be executed locally, at an edge/MEC node, or in the cloud. This flexibility creates a scheduling problem in which the system must select an execution location and allocate resources while responding to changing workload and network conditions.

Task scheduling in this environment is difficult because devices and servers are heterogeneous, tasks may arrive unpredictably, and the state of CPU, memory, bandwidth, queues, and connectivity changes over time. A scheduling decision that is efficient at one instant may become unsuitable after congestion, node failure, or a change in task sensitivity. Surveys of edge resource scheduling and reinforcement-learning-based offloading consequently emphasize dynamic decision making, resource allocation, and the need to handle heterogeneous environments [3]–[5].

Security and privacy add another dimension. Offloading can expose sensitive inputs, intermediate data, task characteristics, or user preferences to remote resources. Recent work has therefore considered personalized privacy levels, privacy entropy, and secure offloading constraints alongside latency and energy [6], [7]. Security can also affect the trustworthiness of the resource itself: a highly available node may still be unsuitable for a sensitive task if authentication anomalies, suspicious communication, or historical incidents indicate elevated risk.

Digital Twin technology offers a useful state-management layer. A DT can represent the current or recent state of physical or computational entities and can be updated from real-time observations [8]. In edge environments, the twin can model MEC nodes, cloud resources, network conditions, task states, and selected security indicators. Recent studies use DTs to support data acquisition for learning, task offloading, scheduling, and multi-objective optimization [9]–[12].

Accordingly, secure scheduling should treat performance and security as coupled decision factors rather than independent stages. The motivation for a Local–MEC–Cloud strategy is not simply to select the fastest resource; it is to select a feasible resource that balances latency, energy, resource utilization, deadline constraints, privacy requirements, and risk. This survey reviews the literature around these dimensions, organizes the approaches into a common taxonomy, compares representative studies, identifies research gaps, and derives a potential integrated research direction.

2. LITERATURE OVERVIEW

Edge task scheduling. Traditional approaches such as FCFS, Round Robin, greedy selection, and heuristic scheduling remain useful because they are simple and inexpensive. However, their fixed decision rules can be less responsive to highly dynamic resource states. Distributed resource scheduling surveys identify scalability, low latency, and autonomous decision making as recurring requirements [3]. Recent work also considers dependent tasks, dynamic arrivals, load balancing, and multi-resource queues rather than treating each task as an isolated job [13], [14].

Mobile Edge Computing and learning-based offloading. MEC research has progressively moved from analytical offloading models toward learning-based decision systems. Reinforcement learning is attractive because it can map observed system states to actions without requiring a closed-form solution for every dynamic condition. Comprehensive surveys published in 2023–2024 categorize RL and DRL methods by objective, state/action design, resource allocation, and deployment scenario [4], [5]. Joint DNN partitioning and offloading studies further illustrate how learning can coordinate where computation is split and executed [15].

Digital Twin-enabled scheduling. DTs are increasingly used as a bridge between physical environments and computational decision models. A 2025 survey describes DT-empowered intelligent offloading as a means to improve situational awareness and resource allocation in edge systems [9]. Recent application studies include DT-assisted vehicular cloud-edge offloading with improved A3C [10], DT-based multi-objective scheduling for vehicular edge

networks [11], and DT-enabled AoI-aware scheduling in Industrial IoT edge networks [12]. These studies demonstrate the usefulness of real-time or near-real-time state representation, while also revealing that many formulations remain application-specific.

Optimization and Genetic Algorithms. Scheduling and offloading are frequently formulated as non-convex or NP-hard problems. Genetic Algorithms (GAs) are therefore used to explore large solution spaces through population initialization, selection, crossover, mutation, and fitness evaluation. Recent edge–cloud work uses GA to address heterogeneous resources and deadline constraints [16], while other studies combine GA with directed search or local improvement to accelerate convergence [17]. DT-enabled scheduling has also been combined with GA to optimize AoI and task completion time [12].

Privacy and security-aware scheduling. Privacy-sensitive offloading research has introduced explicit privacy levels or privacy-preservation objectives rather than treating all tasks uniformly [6], [7]. Secure DT-edge research has additionally examined the integrity of scheduling decisions and trust in distributed environments [18]. These studies indicate that security constraints can alter the feasible resource set and therefore should be available to the scheduler before assignment rather than only after execution.

Multi-objective scheduling. Several studies demonstrate that latency-only optimization is insufficient for heterogeneous edge–cloud environments. Multi-objective formulations can include energy, latency, queueing delay, accuracy, load balance, and deadline satisfaction [11], [19]. The literature therefore suggests a transition from single-metric offloading toward broader objective functions, although the security/privacy dimension is still less consistently integrated.

3. METHODOLOGIES AND APPROACHES

Traditional scheduling uses deterministic rules such as FCFS, Round Robin, greedy ranking, or heuristic priority. Its main advantage is low computational overhead and predictable behavior; its limitation is weak adaptation to changing conditions. Optimization-based scheduling formulates an explicit objective and searches a solution space using GA, particle-based methods, or other metaheuristics. Such methods can handle combinatorial decisions but may incur convergence and runtime overhead.

AI-based scheduling learns a mapping between environment state and scheduling decisions. Supervised models can estimate workload or risk, while reinforcement learning and DRL learn policies from interactions. RL methods are useful when the environment is dynamic, but training data, stability, generalization, and online inference overhead remain important concerns [4], [5].

Privacy-aware scheduling explicitly models data sensitivity or privacy requirements. Security-aware scheduling adds trust, authentication, anomaly indicators, or security penalties. DT-based scheduling uses a virtual representation to supply state information, forecast changes, or test decisions before deployment. Multi-objective scheduling combines several performance or security goals, while hybrid AI + optimization approaches separate prediction from search: an AI model estimates state/risk and an optimizer selects a resource assignment.

Figure 1 summarizes this approach. The categories overlap rather than form mutually exclusive groups. For example, a DT-assisted scheduler can use DRL, while a security-aware scheduler can use GA. The important design question is therefore how the components interact and which information is available at decision time.

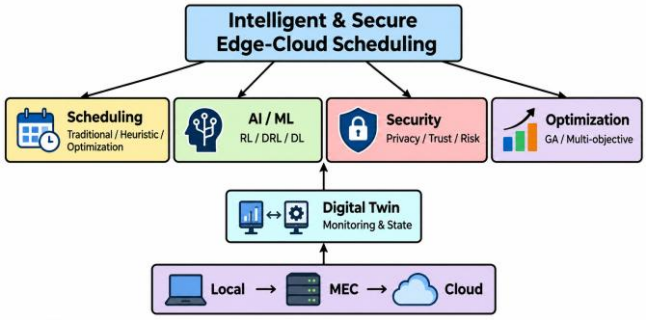


Figure 1. Approaches of Intelligent and Secure Edge–Cloud Task Scheduling.

4. COMPARATIVE LITERATURE ANALYSIS

Table 1 compares representative studies across method, optimization objective, AI/optimization technique, security or privacy treatment, DT usage, environment, and limitations. The comparison shows that current research has strong coverage of performance-oriented offloading and growing coverage of DT-assisted scheduling. However, the combination of real-time DT state, explicit security risk, privacy constraints, and multi-objective Local–MEC–Cloud assignment is less consistently addressed in one framework.

The reviewed studies also differ in evaluation environments. Vehicular studies commonly use SUMO, OMNeT++, and Veins; MEC simulations use synthetic workloads and resource models; security datasets such as CIC-IDS2017 and UNSW-NB15 provide labeled network behavior but do not directly reproduce a DT scheduling environment [20], [21]. This distinction is important when constructing an evaluation methodology.

Study	Year	Method	Objective	AI/Opt.	Security/Privacy	DT	Limitation
Mao et al. [1]	2017	MEC survey	Latency/resource mgmt.	Analytical	Privacy-aware discussion	No	Broad survey
Sabni et al. [3]	2022	Resource scheduling survey	Distributed scheduling	Heuristic/opt.	Limited	No	General edge focus
RL survey [4]	2023	RL offloading survey	Offloading/resource allocation	RL/DRL	Scenario dependent	No	Training overhead
Zhang et al. [15]	2023	DNN partition + offload	Latency/energy	DRL	Not central	No	DNN-specific
Bozkaya et al. [18]	2023	DT-edge scheduling	Energy/delay/integrity	GA/WLP	Integrity	Yes	Blockchain layer overhead
Wu et al. [6]	2024	Privacy-aware offload	QoS + privacy	MADRL	Privacy entropy	No	Complex multi-user setting
Zhu & Tan [10]	2024	DT vehicular offload	Delay/energy	Improved A3C	Not central	Yes	Vehicular scenario
GA skew mutation [16]	2024	GA scheduling	Completion/resource use	GA	No	No	Security not modeled
DT multi-objective [11]	2025	DT + A3C	Latency/energy/accuracy	DRL	Not central	Yes	Vehicular/DNN focus
Bozkaya-Aras [12]	2025	DT + AoI scheduler	AoI/completion time	GA	No	Yes	IIoT-specific

Table 1. Comparative Analysis of Representative Research.

5. FINDINGS AND RESEARCH TRENDS

Five broad trends emerge. First, the architecture is moving from cloud-only execution toward edge–cloud collaboration. Second, DTs are increasingly used to maintain state information and improve scheduling awareness. Third, AI methods are being used for dynamic decisions where analytical optimization is difficult. Fourth, multi-objective optimization is becoming more common because latency, energy, queueing, and resource utilization interact. Fifth, privacy and security are gradually entering the scheduling objective rather than being treated only as post-decision controls.

At the same time, the literature often focuses on a subset of the complete problem. Some approaches optimize latency and energy but omit security; others address privacy but use a restricted execution environment; DT-based approaches may optimize scheduling without modeling adversarial events. Recent research already combines several dimensions—for example, DT, multi-objective scheduling, and DRL [11], or privacy-aware DRL offloading [6]—so the gap should be stated as an integration opportunity rather than a claim that no combined approach exists.

Figure 2 presents a conceptual architecture in which tasks can remain local or move to MEC/cloud resources. The DT continuously represents state information, while a risk engine supplies a security signal to the scheduler. This arrangement separates state observation, risk estimation, and optimization while allowing them to interact.

6. CHALLENGES AND RESEARCH GAPS

Dynamic task arrivals and heterogeneous resources make static schedules fragile. A scheduler must accommodate changes in CPU capacity, memory, bandwidth, queue length, node availability, and task deadlines. Real-time DT synchronization also introduces overhead: the twin is useful only if its state is sufficiently current for the decision being made.

Security-aware scheduling faces an additional modeling challenge. A risk score must be interpretable enough to influence resource selection without becoming a single opaque decision. Candidate inputs can include CPU and memory utilization, task sensitivity, authentication status, historical security events, trust score, communication anomalies, and task type. A lightweight classifier such as Random Forest,

Decision Tree, Gradient Boosting, or a compact neural network could map these features to Low, Medium, or High risk. Such a model should be evaluated independently; no accuracy should be assumed before experiments.

Privacy-preserving offloading must balance protection with computational cost. Over-protection can increase latency or energy consumption, while under-protection can expose sensitive tasks. Failure recovery and adaptive re-scheduling are similarly important because an edge node may become unavailable after assignment. Dataset limitations are also significant: cybersecurity datasets describe traffic or attack behavior, whereas scheduling datasets describe resource states and task execution. Combining them requires careful feature alignment rather than assuming they are interchangeable.

Table 2 summarizes these gaps. The central opportunity is an integrated framework that connects DT state representation, AI risk assessment, privacy awareness, GA-based multi-objective optimization, Local–MEC–Cloud selection, and dynamic adaptation.

Existing Approach	Observed Limitation	Research Gap	Potential Direction
Latency/energy scheduling	Security may be external	Joint performance-risk objective	Risk-aware fitness
Privacy-aware offloading	Protection can add cost	Adaptive privacy-performance balance	Sensitivity-aware penalties
DT scheduling	State freshness overhead	Real-time synchronization policy	Age/freshness-aware DT
AI-only scheduling	Training/generalization cost	Transparent optimization layer	AI risk + GA search
GA scheduling	May ignore security context	Security-aware chromosome fitness	Risk-constrained GA
Static offloading	Cannot react to failures	Continuous adaptation	Trigger-based re-scheduling
Cyber datasets	Not DT scheduling benchmarks	Feature/domain mismatch	Hybrid dataset + simulation
Single-layer execution	Limited flexibility	Local–MEC–Cloud selection	Three-tier candidate discovery

Table 2. Major Research Gaps and Potential Research Directions.

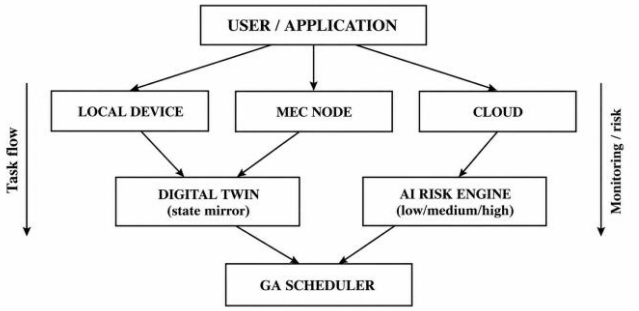


Figure 2. Digital Twin-Enabled Local–MEC–Cloud Architecture.

7. PROPOSED RESEARCH DIRECTION

Based on the reviewed gaps, a potential research direction is a Secure and Intelligent Multi-Objective Task Scheduling framework for Digital Twin-enabled Local–MEC–Cloud networks. This is a proposed research direction, not a claim of completed experimental validation.

The architecture can contain ten logical components: task arrival, task classification, DT monitoring, AI-based risk assessment, candidate resource selection, GA scheduling, secure task assignment, task execution, monitoring, and adaptive re-scheduling. Task classification can identify computational intensity, deadline, data sensitivity, and security requirements. The DT supplies current resource and network information. The risk engine estimates whether a resource/task combination should be considered low, medium, or high risk. The GA then searches feasible assignments.

The AI risk model and GA have different responsibilities. The risk model estimates security characteristics; it does not need to solve the complete scheduling problem. The GA represents candidate allocations as chromosomes, evaluates a fitness function, selects promising candidates, applies crossover and mutation, and terminates after a stopping condition. This separation makes the design easier to analyze and permits different risk models or optimization algorithms to be substituted.

A conceptual multi-objective fitness function is $F = w1L + w2E + w3P + w4R + w5D$, where L is

latency, E is energy consumption, P is privacy/security penalty, R is resource utilization or risk, and D is deadline violation/delay penalty. The weights are application-specific. The formula is a research design abstraction and should be calibrated and validated experimentally rather than treated as an established result.

Adaptive scheduling is essential because scheduling should not be a one-time decision. The framework can trigger re-evaluation when new tasks arrive, resources change, network conditions deteriorate, security risk changes, an MEC node fails, latency increases, or a resource becomes overloaded. Figure 3 shows the resulting closed-loop workflow.

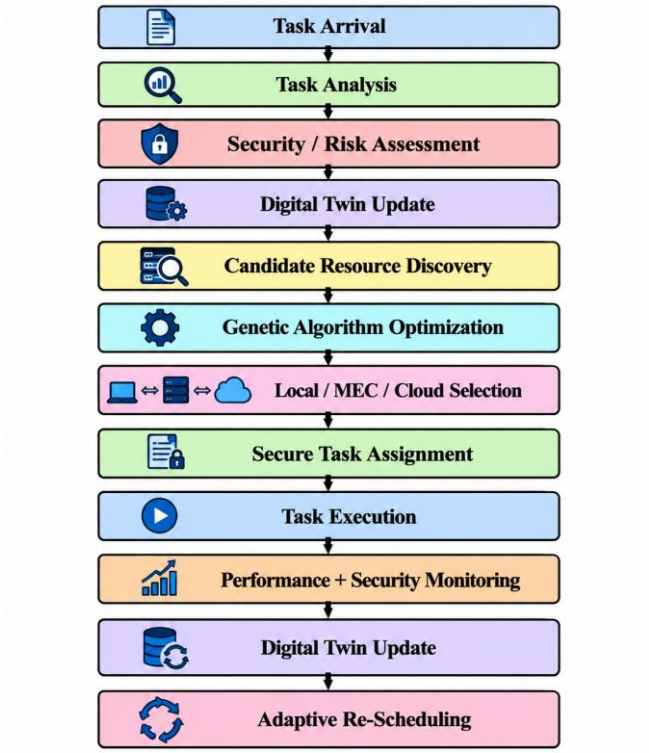


Figure 3. Secure Adaptive Task Scheduling Workflow.

8. EVALUATION CONSIDERATIONS

Evaluation should compare the proposed direction with simple and optimization-based baselines such as FCFS, Round Robin, greedy scheduling, random scheduling, traditional GA, AI-only scheduling, and security-unaware scheduling. Numerical improvement should be reported only after actual experiments. A useful evaluation matrix should separate performance, resource, scheduling, security, privacy, and reliability metrics.

Performance metrics include task completion time, end-to-end latency, throughput, and deadline satisfaction. Resource metrics include CPU utilization, memory utilization, energy consumption, and overall resource utilization. Scheduling metrics include success rate, scheduling overhead, queue waiting time, and load balance. Security metrics include risk-classification performance, attack detection rate, false positive rate, false negative rate, and security violations. Privacy metrics can include privacy penalty and sensitive-task exposure. Reliability metrics include task failure rate, recovery time, and re-scheduling time.

Controlled attack simulation can be used to test whether the scheduler reacts appropriately to abnormal traffic, authentication failures, resource exhaustion, suspicious task requests, communication anomalies, and service disruption. These should be simulated in a controlled environment; the purpose is to evaluate scheduling resilience, not to attack real systems. Public datasets such as CIC-IDS2017, UNSW-NB15, TON_IoT, and Edge-IIoTset can support security-model development, but they should be described accurately as cybersecurity datasets rather than direct DT scheduling benchmarks [20]–[23].

Table 3 provides a compact evaluation framework. A strong study should report not only average values but also distributions, workload sensitivity, scalability, and the cost of re-scheduling. Ablation studies can isolate the contribution of the DT, risk model, privacy term, and GA optimizer.

Category	Metric	Purpose
Performance	Task completion time	Measure end-to-end execution cost
Performance	Latency / throughput	Assess responsiveness and service capacity
Resource	CPU / memory utilization	Measure resource efficiency
Resource	Energy consumption	Assess device/edge energy cost
Scheduling	Success rate / queue time	Measure scheduling effectiveness
Scheduling	Overhead / load balance	Assess scheduler cost and distribution

Security	Risk accuracy / FPR / FNR	Evaluate risk model behavior
Security	Security violations	Measure unsafe assignments
Privacy	Privacy penalty / exposure	Quantify sensitivity-aware protection
Reliability	Failure / recovery time	Evaluate resilience and re-scheduling

Table 3. Evaluation Metrics for Secure Edge–Cloud Scheduling.

9. CONCLUSION

Edge–cloud task scheduling is evolving from simple offloading decisions toward adaptive, multi-objective resource management. The reviewed literature demonstrates active research in MEC scheduling, reinforcement learning, DT-assisted offloading, privacy-aware computation, and genetic optimization. DTs are particularly relevant because they can provide a continuously updated representation of computing and network state, while AI and optimization methods can transform that information into scheduling decisions.

The survey also indicates that performance and security are often studied with different objectives or in limited combinations. Recent work shows that several combinations are feasible, including DT with DRL, DT with multi-objective optimization, and privacy-aware DRL. The resulting research opportunity is therefore to integrate these dimensions carefully rather than claim that each has been unexplored.

A potential future framework can combine Local–MEC–Cloud execution, DT monitoring, AI-based risk assessment, privacy-aware constraints, GA-based multi-objective optimization, controlled attack simulation, and adaptive re-scheduling. Its value should ultimately be demonstrated through transparent experiments using appropriate baselines, realistic workload changes, and security-aware evaluation. The proposed direction provides a structured basis for such future investigation without presenting unvalidated architecture or numerical performance as established results.

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